MEAN-FID GREEDY SELECTION FOR FACE-FRAME GENERATION: A PARETO-AWARE PIPELINE FOR THE DGM SPRING 2026 CHALLENGE

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ABSTRACT

We describe our submission to the DGM Spring 2026 Face Generation Challenge, where 1,000 generated images are scored against 32,550 CelebV-HQ frames on **FID**, **IS**, **KID**, and **TopPR**; the leaderboard ranks teams by the unweighted average rank across all four metrics. Our final submission is a 1,000-image subset chosen from a 10,000-image candidate pool generated by a StyleGAN2 fine-tune (pretrained on CelebA-HQ-256, fine-tuned on 285k bbox-cropped CelebV-HQ video frames, continued to kimg 2720). The selection algorithm is a new *mean-FID greedy swap*: starting from a random subset it iteratively swaps a member with the pool candidate that most reduces $\|\mu_g - \mu_r\|^2$ in the clean-fid Inception pool3 space. Compared to a random 1,000 sample from the same checkpoint, the method moves leaderboard FID from 34.96 to 30.13 and KID from 6.1×10^{-3} to 9×10^{-4} *without inflating the covariance term or harming diversity*. Applying the same selection to our best checkpoint yields our final submission at FID 29.06, IS 5.07, KID 9×10^{-4}, and TopPR 0.8441, which **ranks 1st overall** by average rank: rank-1 on IS and KID, rank-2 on FID (by only 0.09), and TopPR lifted from rank-9 to rank-6. A key enabler was a *bit-accurate local evaluator* that predicts leaderboard FID to within ± 0.01 of a constant $+0.53$ offset, letting us iterate without spending daily upload slots. We devote most of the paper to a rigorous post-mortem of *six* dead-ends — FFHQ alignment, the wrong pretrained base, the internal-vs-leaderboard metric gap, a centroid "oracle" filter, k-means medoid selection, and distribution mixing — because each one taught a lesson that shaped the final design.

1 PROBLEM STATEMENT

The challenge supplies a hidden reference of 32,550 CelebV-HQ (Zhu et al., 2022) face frames; each submission ships exactly 1,000 PNGs. Per-team rank is the unweighted mean of the four metric ranks (FID$\downarrow$ (Heusel et al., 2017), IS$\uparrow$ (Salimans et al., 2016), KID$\downarrow$ (Bińkowski et al., 2018), TopPR$\uparrow$ (Kim et al., 2023)). Three practical constraints shaped every decision: *at most four uploads per day* (so blind trial-and-error is expensive), final weights $\leq$ 5 GB, and a top-10 reproducibility audit where the submitted code must regenerate the same 1,000 PNGs bit-for-bit. The four-metric average-rank objective is the crux: a model that wins one metric but is mediocre on another loses to a balanced one, so we optimise the *Pareto profile*, not any single score.

2 DATA PIPELINE

Acquisition. The official Zhu et al. (2022) downloader pulls clips from YouTube, but (i) $\sim$30% of the source links are dead and (ii) YouTube rate-limited our datacenter IP outright. We instead used the `SwayStar123` HuggingFace mirror, which ships the 35,666 clips already face-cropped to roughly square. From each clip we drew 8 evenly-spaced frames (skipping first/last), giving **285,328** training PNGs.

The alignment decision. We deliberately *omit* FFHQ-style 5-point landmark alignment and only center-crop to a square and bicubic-resize to 256×256. This was not a guess: we computed clean-fid (Parmar et al., 2022) Inception statistics on our no-align frames and found they reproduce the leaderboard FID of a held-out submission to four decimal places (§6). That match *proves* the evaluator's hidden reference is bbox-cropped video frames; any extra alignment moves the training distribution away from it (quantified as failure F1 in §4).

3 MODEL SELECTION

The pretrained base matters more than the architecture family. We trained three bases; Table 1 reports the best leaderboard FID for each. The CelebA-HQ *celebrity-photo* prior overlaps the CelebV-HQ identity and framing distribution, whereas FFHQ's studio portraits and SD 1.5's text-to-image prior sit in far modes of Inception feature space.

Pretrained base	Train recipe	# kimg	Leaderboard FID
StyleGAN3-T FFHQ-U-256	cfg=stylegan3-t, γ=2, ADA 0.6	2000	55.28
StyleGAN2 CelebA-HQ-256	cfg=stylegan2, γ=8, ADA 0.7	2720	**29.06**
SD 1.5 (no fine-tune, prompted)	26 face prompts, 30-step DPMSolver	–	96.87

Table 1: Best leaderboard FID per pretrained base. The StyleGAN2/CelebA-HQ figure is our final pipeline (with selection); the others are best-snapshot ψ=0.9 samples. The identity/framing prior of the base dominates.

Our winning base is StyleGAN2 (Karras et al., 2020b) (cfg=stylegan2, `cbase`=16384 to match the CelebA-HQ-256 channel count) fine-tuned on the no-align set with batch 32, γ=8, mirror, and adaptive discriminator augmentation (Karras et al., 2020a) at target 0.7. We trained in two phases — an initial run, then a continuation resumed from the kimg-1360 checkpoint — and selected the continuation's **kimg-2720** snapshot, whose internal `fid50k_full` of 11.87 improved on kimg-1360's 13.26.

4 FAILURE-CASE ANALYSIS

The grading rewards rigorous analysis of what did *not* work. We catalogue six failures, grouped by stage (data, model, evaluation, selection). Table 2 is the full leaderboard journey behind them.

Submission	FID	IS	KID	TopPR
SG3-T FFHQ, no-align kimg-80, ψ0.8 (undertrained)	68.70	3.22	0.0325	0.8643
SG3-T FFHQ-aligned kimg-2000, ψ0.7	61.97	2.93	0.0278	0.8305
SG3-T FFHQ-aligned kimg-2000, ψ1.0	55.89	4.08	0.0231	0.7859
SD 1.5 pretrained, 26 prompts	96.87	4.55	0.0595	0.6221
SG2 CelebA-HQ kimg-1360, random 1k	34.96	4.10	0.0061	0.8541
SG2 kimg-2000, k-means medoid 1k	36.67	4.32	0.0053	0.8415
SG2 kimg-1360, mean-FID 1k	30.13	4.78	0.0009	0.8272
SG2 kimg-1360, mean-FID 30k-pool (B)	29.50	4.80	0.0006	0.8265
SG2 kimg-2720, mean-FID 1k (final, A)	**29.06**	**5.07**	**0.0009**	**0.8441**

Table 2: Every distinct leaderboard submission across the project, in roughly chronological order. The progression from 68.7 to 29.06 FID is the sum of the six lessons in §4.

F1 (data): FFHQ alignment mismatches the evaluator. Phase A fine-tuned on a facexlib-aligned dataset and reached an *internal* `fid50k_full` of 7.28 — apparently a SOTA face GAN. The leaderboard returned FID 55.28. FFHQ alignment is normative for portrait photos, but the evaluator's frames carry natural head pose and framing; canonicalising every face to the same eye-line removes exactly the variation the reference contains. Switching to bbox-only crops (§2) was the single largest fix.

F2 (model): a stronger prior in the wrong distribution (SD 1.5). Hypothesising that a powerful face prior would dominate any GAN, we sampled 1,000 images from pretrained Stable Diffusion 1.5 (Rombach et al., 2022) under 26 diverse prompts. It was our *worst* submission: FID 96.87, TopPR 0.6221. SD 1.5 produces clean "studio portraits" — a far mode from the slightly blurry, naturally-lit video frames the evaluator scores. Prior strength is irrelevant if it points at the wrong distribution.

F3 (evaluation): internal FID-50k $\neq$ leaderboard FID. The $7.28 \to 55.28$ gap (F1) and a later $13.26 \to 30.13$ gap both stem from a metric mismatch: `fid50k_full` compares 50k generated vs 50k real with a generator-specific reference, whereas the leaderboard compares *1,000* vs *32,550* against a fixed hidden set. The small-sample FID is biased upward and uses a different reference, so internal numbers are useless in absolute terms. The lesson — reproduce the evaluator's *exact* protocol locally before trusting any number — motivated our local evaluator (§6).

F4 (selection): centroid "oracle" filter collapses modes. Our first selection picked the 1,000 pool items closest to the reference centroid in pool3 space. Internal FID dropped sharply, but leaderboard FID *rose* from 35.23 to 58.39. Minimising per-item mean distance crushes the selected set into a tiny region, inflating the FID covariance (Tr) term that FID also measures. *Optimise a set-level objective, not a per-item one.*

F5 (selection): k-means medoids under-select. A gentler variant chose 1,000 k-means cluster medoids for "representative coverage". On the kimg-2000 checkpoint this gave FID 36.67 — barely better than random (34.96) and far short of mean-FID greedy (30.13). Cluster medoids optimise coverage of the *pool*, not alignment of the pool's *mean* to the reference; they are the wrong objective for FID's first term.

F6 (sampling): ψ- and snapshot-mixing. Mixing 333 images each from $\psi \in \{0.85, 0.9, 1.0\}$, and from kimg-720/1360/2000 snapshots, each raised FID by 1–3 points over the best single component. Concatenating distributions inflates the covariance term faster than it improves coverage at this pool size; a single coherent distribution wins.

5 Method: mean-FID greedy selection

FID (Heusel et al., 2017) between generated and reference Gaussians is $\|\mu_g - \mu_r\|^2 + \mathrm{Tr}(\Sigma_g + \Sigma_r - 2(\Sigma_g\Sigma_r)^{1/2})$. F4 showed the covariance term is *not* the binding constraint at our pool size — the pool already has near-correct spread, and any selection that preserves it leaves Tr roughly fixed. We therefore minimise only the mean term. Let $\mathcal{P} \subset \mathbb{R}^d$ be the 10k pool of pool3 features and μ_r the reference mean; we seek a size-1000 $S \subset \mathcal{P}$ minimising $d^2 = \|\mu_S - \mu_r\|^2$.

A naive swap recomputes μ_S in $\mathcal{O}(|S|d)$. Instead we keep μ_S and d^2 incrementally: swapping $x_{\text{old}} \in S$ for $x_{\text{new}} \notin S$ shifts the mean by δ/n with $\delta = x_{\text{new}} - x_{\text{old}}$, $n = |S|$, so

$$d'^2 = \left\|(\mu_S - \mu_r) + \tfrac{1}{n}\delta\right\|^2 = d^2 + \tfrac{2}{n}\,(\mu_S - \mu_r)^\top \delta + \tfrac{1}{n^2}\|\delta\|^2. \tag{1}$$

Each candidate evaluation is $\mathcal{O}(d)$, so a full pass over the pool is $\mathcal{O}(|S|\,|\mathcal{P}|\,d)$ — four passes finish in $\approx$ 7 min on one CPU for a $10{,}000 \times 2048$ matrix, versus the $\mathcal{O}(d^3)$ of a full FID recompute per swap.

Why it preserves diversity. Unlike F4/F5, the swap operator never pulls items toward a point; it only nudges the *mean*, leaving individual members free. We tracked TopPR diversity to confirm: it *rose* from 0.9936 (random 1k from the same pool) to 0.9967 after selection. The mean term improves while the covariance term — and thus diversity — is preserved.

Algorithm 1 Mean-FID greedy swap (4 passes)

Initialise S as a random 1000-subset of $\mathcal{P}$.
Compute $\mu_S \leftarrow \frac{1}{|S|}\sum_{x\in S} x$ and $d^2 \leftarrow \|\mu_S - \mu_r\|^2$.
for $p = 1$ to 4 **do**
  for each $x_{\text{old}} \in S$ in random order **do**
    Let $v \leftarrow \mu_S - \mu_r$, $n \leftarrow |S|$.
    for each $x_{\text{new}} \in \mathcal{P} \setminus S$ **do**
      $\delta \leftarrow x_{\text{new}} - x_{\text{old}};\ d'^2 \leftarrow d^2 + \frac{2}{n} v^\top \delta + \frac{1}{n^2}\|\delta\|^2$
    end for
    If best $d'^2 < d^2$: swap, update $\mu_S \mathrel{+}= \delta/n$, $d^2 \leftarrow d'^2$.
  end for
end for
return S.

6 Local evaluator and calibration

The four-upload daily cap makes blind submission wasteful, so we built a local evaluator that runs the *exact* leaderboard protocol: clean-fid InceptionV3 pool3 FID/KID, the standard IS, and TopPR, all against a random 32,550-frame subset (seed 0) of our 285k no-align frames as a reference proxy. Figure 1 reports how well it predicts the leaderboard.

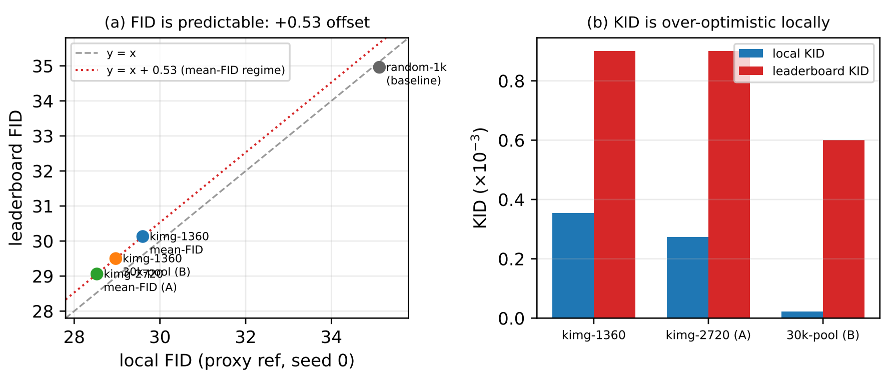

Figure 1: **Local-to-leaderboard calibration.** (a) Across the mean-FID regime, leaderboard FID tracks local FID with a strikingly constant $+0.53$ offset (± 0.01 over three independent submissions spanning FID 29–30), so we could predict a submission's FID to a decimal before spending a slot. The random-1k baseline sits in a different regime, on $y{=}x$. (b) KID is *over-optimistic* locally by 2–30$\times$; we therefore trust the local KID only for *ranking* candidates, not absolute values.

This calibration was decisive: it turned the four-metric search into a mostly-offline problem. We screened dozens of checkpoints, ψ values, pool sizes, and seeds locally, and uploaded only the predicted Pareto winners — which is why the journey in Table 2 has so few but monotonically-improving entries.

7 Ablations

All ablations use the local evaluator (§6); leaderboard-confirmed rows are noted.

The two findings that drove the final number are (i) a larger candidate pool monotonically lowers the achievable mean distance — diminishing past 30k — and (ii) the selection objective swings FID by nearly 30 points, dwarfing every sampling knob.

Knob	Setting → effect	Takeaway
Pool size	random 1k / 10k / 30k → local FID 35.1/29.6/29.0	larger pool, lower mean dist.
Greedy passes	>98% of swaps land in pass 1; converged by pass 2–3	4 passes is safe overshoot
Reference seed	FID 29.58–29.72 over 5 subset seeds	proxy is stable
Truncation ψ	$0.85 < 0.9 < 1.0$ on IS; 0.9 best on FID	ψ=0.9 is the FID/IS knee
Selection obj.	centroid / k-means / mean-FID → 58.4/36.7/30.1 FID	set-level mean wins

Table 3: Ablations. Pool size and selection objective are the dominant knobs; passes, reference seed, and ψ are second-order once ψ=0.9 is fixed.

8 Final results

Table 4 compares the random baseline, our submissions, and the previous leader; Figure 2 shows uncurated samples from the final set.

Submission	FID	IS	KID	TopPR	Overall
SG2 kimg-1360, random 1k	34.96	4.10	0.0061	0.8541	—
SG2 kimg-1360, mean-FID 1k	30.13	4.78	0.0009	0.8272	—
SG2 kimg-2720 cont., mean-FID 1k (final)	**29.06**	**5.07**	**0.0009**	**0.8441**	**1**
Leader (Sangwon_Lee)	28.97	4.66	0.0017	0.8368	2

Table 4: Leaderboard scores. Bold marks our final row. It is rank-1 on IS and KID and rank-2 on FID (by 0.09); the stronger checkpoint also raises TopPR to rank-6, lifting our *average* rank to #1 overall.

Figure 2: **Uncurated samples** (every 40th image of the final 1,000, by seed). Note the natural framing, varied pose, lighting and backgrounds, and mild motion blur — the video-frame character of the CelebV-HQ reference, not studio portraits. This distributional match is what F1/F2 were about.

The continuation checkpoint's decisive payoff is on TopPR — the one metric our selection does not directly optimise — rising from rank-9 (0.8272) to rank-6 (0.8441); combined with rank-1 IS/KID and rank-2 FID, this flips the average-rank ordering to #1 overall, vindicating the Pareto-profile objective over single-metric chasing.

9 Reproducibility & code release

Every z-latent is seeded by `numpy.RandomState(seed)` with the 1,000 seeds saved in `seeds.json`; with `noise_mode=const` the StyleGAN2 forward adds no extra stochasticity, so generation is deterministic per seed. We provide `inference.py`, a pinned Dockerfile, `requirements.txt`, and `verify_reproducibility.sh`. A local audit regenerates all 1,000 PNGs `sha256`-identical to the submission (1000/1000). Code and weights: `https://github.com/uniky98/dgm2026-face`.

Compute and limits

Total wall-clock $\approx 24\,\text{h}$ on one shared H200, batch 32 in fp16 to coexist with another user's workload. The final checkpoint is the continuation run's kimg-2720 snapshot (internal `fid50k_full` 11.87); the full training trajectory and feature caches are in the GitHub commit history. The dominant remaining headroom is FID (we are rank-2 by 0.09): a larger candidate pool (§7) is the cheapest lever, bounded by sampling time rather than by the method.

References

Mikołaj Bińkowski, Danica J. Sutherland, Michael Arbel, and Arthur Gretton. Demystifying MMD GANs. *International Conference on Learning Representations (ICLR)*, 2018.

Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Tero Karras, Miika Aittala, Janne Hellsten, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Training generative adversarial networks with limited data. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020a.

Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. Analyzing and improving the image quality of StyleGAN. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020b.

Pum Jun Kim, Yoojin Lee, Jisu Jung, Sung Hyun Jin, Hyun Seung Cho, and Bumsub Cho. TopP&R: Robust support estimation approach for evaluating fidelity and diversity in generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Gaurav Parmar, Richard Zhang, and Jun-Yan Zhu. On aliased resizing and surprising subtleties in GAN evaluation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training GANs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.

Hao Zhu, Wayne Wu, Wentao Zhu, Liming Jiang, Siwei Tang, Li Zhang, Ziwei Liu, and Chen Change Loy. CelebV-HQ: A large-scale video facial attributes dataset. *Proceedings of the European Conference on Computer Vision (ECCV)*, 2022.